

Python Full Stack

Professional Program – 240 Hours

Course Description

This course equips learners with comprehensive skills in Python Full Stack Development, combining frontend (HTML5, CSS3, JavaScript, ReactJS), backend (Python, Django/Flask with REST APIs), and databases (MySQL, MongoDB). The program emphasizes practical, project-based learning, ensuring participants can design, build, and deploy scalable full-stack applications.

Module 1: Python Programming Fundamentals (30 Hours)

Syllabus Topics

- What is a program, memory, and languages
- Python Introduction, Features & Versions
- Data Types, Variables, Operators
- Control Structures (if, loops, switch)
- Functions & Recursion
- Lists, Tuples, Sets, Dictionaries
- Strings & File Handling

Practicals

- Write Python programs for arithmetic & data structures
- File read/write mini-project
- Build a simple CLI-based calculator

Tools

- Python 3.x
 - Jupyter Notebook
 - PyCharm/VS Code
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Module 2: Advanced Python & OOP (30 Hours)

Syllabus Topics

- Exception Handling
- OOP Principles: Classes, Objects, Inheritance, Polymorphism, Encapsulation
- Modules & Packages
- Iterators & Generators
- Decorators

- Python Libraries (OS, Sys, Math, DateTime)

Practicals

- Implement OOP mini-project (Banking / Student System)
- Exception-handling exercises
- Create Python modules and packages

Tools

- Python
 - PyCharm
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Module 3: MySQL Database (30 Hours)

Syllabus Topics

- DBMS & RDBMS Concepts
- MySQL Features
- ER Diagrams & Normalization
- SQL Commands: DDL, DML, DCL, TCL
- Queries: SELECT, WHERE, LIKE, ORDER BY, BETWEEN
- Joins, Subqueries, Views, Stored Procedures

Practicals

- Create and query a relational schema
- Write business-related queries (top sales, active users)
- Implement stored procedures

Tools

- MySQL
 - MySQL Workbench
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Module 4: MongoDB Database (25 Hours)

Syllabus Topics

- Introduction to NoSQL Databases
- MongoDB Installation & Compass
- JSON vs BSON
- CRUD Operations
- Collections & Documents
- Aggregation Framework & Indexing

Practicals

- Create collections and perform CRUD

- Design schema for shopping cart system
- Query data using aggregations

Tools

- MongoDB
 - MongoDB Atlas
 - Mongoose
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Module 5: HTML5 (20 Hours)

Syllabus Topics

- Introduction to HTML & Elements
- Headings, Paragraphs, Links, Lists
- Tables, Forms, Multimedia
- Semantic Elements & Attributes

Practicals

- Build a personal portfolio site
- Create form-based input page
- Develop a multi-page static website

Tools

- VS Code
 - Chrome DevTools
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Module 6: CSS3 (25 Hours)

Syllabus Topics

- Introduction & Syntax
- Inline, Internal, External CSS
- Selectors & Rules
- Flexbox & Grid
- Responsive Design with Media Queries
- Bootstrap Integration

Practicals

- Apply CSS styling to web forms
- Build responsive landing page
- Style dashboard with Flexbox/Grid

Tools

- VS Code

- Bootstrap
- Material UI

Module 7: JavaScript & ES6+ (30 Hours)

Syllabus Topics

- JavaScript Basics: Syntax, Variables, Functions
- DOM Manipulation
- Events & Form Validation
- ES6 Features (Arrow Functions, Classes, Modules)
- Async Programming (Callbacks, Promises, Async/Await, Fetch API)

Practicals

- Create a JavaScript-based form validator
- Fetch API for live data
- Build mini To-Do app

Tools

- Node.js
- VS Code
- Chrome DevTools

Module 8: ReactJS Front-End (40 Hours)

Syllabus Topics

- Introduction & Setup
- Components, JSX, Props, State
- Event Handling & Lifecycle
- React Hooks (useState, useEffect)
- Routing with React Router
- State Management with Redux

Practicals

- Build Todo app with React
- CRUD app with REST API integration
- Implement Redux store for global state

Tools

- ReactJS
- Redux

- React Router

- Material UI

Module 9: Django/Flask Backend Development (40 Hours)

Syllabus Topics

- Django/Flask Basics
- Project Setup & Architecture
- Models, Views, Templates
- REST APIs with Django Rest Framework / Flask-RESTful
- ORM Integration with MySQL/MongoDB
- Middleware, Authentication & Authorization

Practicals

- Build REST API for user management
- Integrate Django backend with React frontend
- Implement authentication & role-based access

Tools

- Django
- Django Rest Framework / Flask
- Postman

Module 10: Security, Deployment & Capstone Project (30 Hours)

Syllabus Topics

- JWT Authentication
- Dockerization & CI/CD Basics
- Deployment on AWS/Heroku
- Agile Project Development & Documentation

Practicals

- E-commerce App (cart, payments, auth)
- Social Media App (posts, likes, comments)
- Task Management System (users, roles, tasks)

Tools

- JWT
 - Docker
 - AWS/Heroku
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Course Outcomes

- Write clean, efficient programs using Python fundamentals & OOP.
- Build web backends using Django/Flask and expose REST APIs.
- Use MySQL and MongoDB for relational and non-relational data storage.
- Create responsive user interfaces using HTML5, CSS3, and JavaScript.
- Develop modern frontends using ReactJS with hooks, routing, and state management.
- Implement authentication & security with JWT and Django Rest Framework.
- Apply DevOps practices (Git, Docker, AWS/Heroku) for deployment.